Tutorial 1
Logic, Reasoning & Proofing Techniques

E0 298 | Linear Algebra and its Applications

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Outline

1. Mathematical Logic

1. 1 Propositional Logic

1. 2 Predicate Logic

2. Proof Techniques

Mathematical Logic

Mathematical logic is the language one uses to construct rigorous mathematical proofs. Any mathematical argument proceeds in a sequence of mathematical statements. Statements are either TRUE or FALSE.

Outline

1. Mathematical Logic

1. 1 Propositional Logic

1. 1. 1 Conjunction

1. 1. 2 Disjunction

1. 1. 3 Negation

1. 1. 4 Implication

1. 1. 5 Equivalance

2. Proof Techniques

Propositional (or 0th-order) Logic (or Calculus)

A proposition is a declarative sentence that is either TRUE or FALSE, but not both. In mathematical logic, propositions are the basic building blocks used to construct more complex logical arguments. Examples of propositions:

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- $2 + 2 = 4$ (TRUE)
- $2 + 2 = 5$ (FALSE)

Conjunction

Definition

If X is a statement and Y is a statement, the statement "X AND Y" is TRUE if X and Y are both TRUE and is FALSE otherwise. AND is represented by the logical operator $\wedge$.

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If X is a statement and Y is a statement, the statement "X AND Y" is TRUE if X and Y are both TRUE and is FALSE otherwise. AND is represented by the logical operator $\wedge$.

Example

1. "$2 + 2 = 4 \wedge 3 + 4 = 7$" is TRUE.
2. "$2 + 2 = 4 \wedge 3 + 4 = 9$" is FALSE.

Disjunction

Definition

If X is a statement and Y is a statement, the statement "X OR Y" is TRUE if either X or Y or both are TRUE, i.e., in mathematical logic, the OR is an *inclusive* OR. OR is represented by the logical operator $\vee$.

Example

1. "$2 + 2 = 4 \vee 3 + 4 = 9$" IS TRUE.
2. "$2 + 2 = 5 \vee 3 + 4 = 9$" IS FALSE.

Negation

Definition

The statement "X IS NOT TRUE" or "X IS FALSE" or simply "NOT X" is called the negation of X, and is TRUE when X is FALSE, and FALSE whenever X is TRUE. Negation or NOT is represented by the logical operator $\neg$. Negations convert AND into OR and vice versa in accordance with the De Morgan's Laws.

Example

1. The statement "$2 + 2 = 5$ is FALSE" is TRUE.
2. The negation of the statement "Jane has black hair and Jane has blue eyes" is "Jane doesn't have black hair or doesn't have blue eyes".

Implication (or Conditional)

Definition

If X is a statement, and Y is a statement, then "if X, then Y" is the implication from X to Y. If X is TRUE, then "if X, then Y" is TRUE when Y is TRUE, and FALSE when Y is FALSE. However, if X is FALSE, then "if X, then Y" is always TRUE (vacuously), irrespective of whether Y is TRUE or FALSE! Implication is represented by the logical operator $\Rightarrow$ or $\rightarrow$.

Example

1. Given $x = 2$. The statement "If $x = 2$, then $x^2 = 4$" is TRUE. (Why?)
2. Given $x = 2$. The statement "If $x = 2$, then $x^2 = 9$" is FALSE. (Why?)
3. Given $x = 3$. The statement "If $x = 2$, then $x^2 = 9$" is TRUE. (Why?)

Implication Expressions

$$p \to q$$

- Hypothesis / Premise / Antecedent: p
- Conclusion / Consequence: q

If p then q

If p is true, then q is true

p implies q

p only if q

q if p

q when p

q follows from p

q whenever p

q unless $\neg p$

A sufficient condition for q is p.

A necessary condition for p is q.

q is necessary for p.

Equivalance (or Bi-conditional)

Definition

If X is a statement, and Y is a statement, we say that "X is TRUE IFF [1]Y is TRUE", whenever X is TRUE, Y has to be TRUE, and whenever Y is TRUE, X has to be TRUE also. Intuitively, both, X and Y, are logically equivalent statements. Equivalance is represented by the logical operator $\Leftrightarrow$ or $\leftrightarrow$.

Example

1. Given $x \in \mathbb{Z}$ the statement "$x = 3$ IFF $x^2 = 9$" is FALSE. (Why?)
2. Given $x \in \mathbb{Z}$ the statement "$x = 3$ IFF $2x = 6$" is TRUE.

[1] read *if and only if*

Outline

1. Mathematical Logic

1. 2 Predicate Logic

1. 2. 1 Existential Quantifier

1. 2. 2 Universal Quantifier

1. 2. 3 Uniqueness Quantifier

1. 2. 4 Negation of quantifiers

1. 2. 5 Combinations of quantifiers

2. Proof Techniques

Predicate (or 1st-order) Logic (or Calculus)

A predicate is an expression of one or more variables determined on some specific domain. A predicate with variables can be made a proposition by either authorizing a value to the variable or by quantifying the variable. Predicates can be seen as boolean functions.

Example

- $E(x, y) :=$ "$x = y$"
- $L(a, b, c) :=$ "$a + b + c = 0$"
- $M(p, q) :=$" p is married to q"

Existential Quantifier $\exists$

Definition

The statement "$P(x)$ is true for *some* x of type T" ($\exists x \in T : P(x)$) means that there exists *at least one* x of type T for which $P(x)$ is true, although it may be that there is more than one such x.

Universal Quantifier $\forall$

Definition

The statement "$P(x)$ is true *for all* x of type T" ($\forall x \in T : P(x)$) means that given *any* x of type T, the statement $P(x)$ is true regardless of what the exact value of x is.

Exercise

Show that $\forall x : P(x) \Rightarrow \exists x : P(x)$

Uniqueness Quantifier $\exists!$

Definition

The statement "$P(x)$ is true for one and only one x of type T" ($\exists! x \in T : P(x)$) means that there exists *exactly one* x of type T for which $P(x)$ is true, i.e., there is no more than one such x.

Exercise

Represent the statement $\exists! x : P(x)$ using only $\forall$ and $\exists$ quantifiers, i.e., show that existensial and universal quantifiers are complete.

Negation of quantifiers

- Negating a universal statement produces an existential statement.
  E.g., Negation of "All swans are white" is not "All swans are not white", but rather "There is some swan which is not white" ;i.e;
  $\neg\forall x : S(x) \Rightarrow W(x) \not\equiv \forall x : S(x) \Rightarrow \neg W(x)$
  $\neg\forall x : S(x) \Rightarrow W(x) \equiv \exists x : S(x) \wedge \neg W(x)$
- Negating an existential statement produces a universal statement.
  E.g., Negation of "There exists a black swan" is not "There exists a swan that is not black", but rather "All swans are not black".
  $\neg\exists x : S(x) \wedge B(x) \not\equiv \exists x : S(x) \wedge \neg B(x)$
  $\neg\exists x : S(x) \wedge B(x) \equiv \forall x : S(x) \Rightarrow \neg B(x)$

1. $\neg(\forall x\, P(x)) \equiv \exists x\, \neg P(x)$
2. $\neg(\exists x\, P(x)) \equiv \forall x\, \neg P(x)$

Example

Let's try to prove (1): $\neg(\forall x\, P(x)) \Leftrightarrow \exists x\, \neg P(x)$

- Claim :- $\neg(\forall x\, P(x)) \Rightarrow \exists x\, \neg P(x)$
  Proof :- LHS:$\neg(\forall x : P(x))$; RHS: $\exists x : (\neg P(x))$
  LHS:$\neg(P_1 \wedge P_2 \wedge P_3 \ldots)$; RHS: $(\neg P_1 \vee \neg P_2 \vee \neg P_3 \ldots)$
  Using De Morgan's Law, we can prove LHS $\Rightarrow$ RHS.
- Claim :- $\exists x : \neg P(x) \Rightarrow \neg(\forall x : P(x))$
  Proof :- RHS:$\neg(\forall x : P(x))$; LHS: $\exists x : (\neg P(x))$
  LHS:$\neg(\neg P_1 \vee \neg P_2 \vee \neg P_3 \ldots)$; RHS: $\neg(P_1 \wedge P_2 \wedge P_3 \ldots)$
  Using De Morgan's Law, we can prove LHS $\Rightarrow$ RHS.

$\therefore \neg(\forall x\, P(x)) \equiv \exists x\, \neg P(x)$

Combinations of quantifiers

- The universal and existential quantifiers can be combined in mathematical statements. Roughly, there are four possible combinations. E.g., "For every positive number x, there exists a positive number y such that $y^2 = x$"; i.e; $\forall x > 0, \exists y > 0 : y^2 = x$
- Swapping quantifiers makes a lot of difference. E.g.,
  1. For every integer n, there exists an integer m which is larger than n. This statement is always true.
     $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z} : m > n$ (Tautology)
  2. There exists an integer m such that m is larger than n for every integer n. This statement is false since you choose m first and then n that falsifies the condition can always be found.
     $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z} : m > n$ (Fallacy)

$$x, y \in \text{set of students in a college}$$
$$F(x, y) : x \text{ and } y \text{ are friends}$$

- $\forall x \forall y\, F(x, y)$: All students in the college are friendly with each other.
- $\exists y \forall x\, F(x, y)$: Some students are friendly with everyone in the college.
- $\forall x \exists y\, F(x, y)$: Every student is friendly with someone in the college.
- $\exists y \exists x\, F(x, y)$: There is at least one pair of friends in the college.

Quantifier Manipulation

$$\forall x \forall y\, P(x, y) \iff \forall y \forall x\, P(x, y)$$

$$\exists x \exists y\, P(x, y) \iff \exists y \exists x\, P(x, y)$$

$$\forall x \exists y\, P(x, y) \Longleftarrow \exists y \forall x\, P(x, y)$$

$$\begin{array}{ccc} & \forall x \forall y & \\ & \Downarrow & \\ \exists y \forall x & & \exists x \forall y \\ \Downarrow & & \Downarrow \\ \forall y \exists x & & \forall x \exists y \\ & \Downarrow & \\ & \exists x \exists y & \end{array}$$

Outline

1. Mathematical Logic

2. Proof Techniques

2. 1 Direct proof

2. 2 Proof by mathematical induction

2. 3 Proof by contraposition

2. 4 Proof by contradiction

2. 5 Proof by exhaustion

2. 6 Proofs involving sets

Outline

1. Mathematical Logic

2. Proof Techniques

2. 1 Direct proof

Direct proof

Example

Given: Suppose $x \in \mathbb{Z}$.

Claim: If x is odd, then x^2 is odd.

Proof: $\exists m \in \mathbb{Z}$ s.t. $x = 2m + 1 \Rightarrow x^2 = (2m+1)^2 = 4m^2 + 4m + 1$ which is the sum of two even numbers and one odd number and is therefore an odd number.

Exercise

1. Suppose $x \in \mathbb{Z}$. If x is even, then $x^2 - 6x + 5$ is odd.
2. Suppose $x, y \in \mathbb{R}$ and $x, y > 0$. If $x < y$ then $\sqrt{x} < \sqrt{y}$.

Outline

1. Mathematical Logic

2. Proof Techniques

2. 2 Proof by mathematical induction

Proof by mathematical induction

Example

Given: $\lambda_0 = 1$, and $\lambda_t = \frac{1+\sqrt{1+4\lambda_{t-1}^2}}{2}$
Claim: $\lambda_t \geq \frac{1+t}{2}$.
Proof: Base case $t = 0$: $\lambda_0 \geq \frac{1}{2}$ Assume the hypothesis is true for $t = k$, where $k \in \mathbb{N}$, i.e., $\lambda_{k-1} \geq \frac{k}{2}$.

$$\Rightarrow 4\lambda_{k-1}^2 \geq k^2 \Rightarrow 1 + 4\lambda_{k-1}^2 \geq 1 + k^2 \geq k^2$$

$$\Rightarrow \sqrt{1 + 4\lambda_{k-1}^2} \geq \sqrt{k^2} = k$$

$$\Rightarrow \lambda_k = \frac{1 + \sqrt{1 + 4\lambda_{k-1}^2}}{2} \Rightarrow \lambda_k \geq \frac{1+k}{2} \qquad \square$$

Exercise

Let $y_1 = 6$, and for each $n \in \mathbb{N}$, define $y_{n+1} = \frac{2y_n - 6}{3}$.

1. Use induction to prove that the sequence satisfies $\forall n \in \mathbb{N} : y_n > -6$.
2. Use induction to show that the sequence $\{y_n\}_{n \in \mathbb{N}}$ is decreasing.

Outline

1. Mathematical Logic

2. Proof Techniques

2. 3 Proof by contraposition

Proof by contraposition

The statement "$P \Rightarrow Q$" is logically equivalent to "$\neg Q \Rightarrow \neg P$". The latter is easier to prove at times.

Example

Given: $x \in \mathbb{Z}$
Claim: If $x^2 - 6x + 5$ is even, then x is odd.
Proof: We prove the contrapositive, i.e., If x is even, then $x^2 - 6x + 5$ is odd. $\exists z \in \mathbb{Z}$ such that $x = 2m$. $\Rightarrow x^2 - 6x + 5 = 4m^2 - 12m + 5$ which is the sum of two even numbers and one odd number and is therefore an odd number.

Exercise

1. Suppose $a, b \in \mathbb{Z}$. If both ab and $a + b$ are even, then both a and b are even.
2. Suppose $x \in \mathbb{R}$. If $x^3 - x > 0$ then $x > -1$.

Outline

1. Mathematical Logic

2. Proof Techniques

2. 4 Proof by contradiction

Proof by contradiction

Given the statement "P" is true, assume that "$\neg P$" is true or P is false and arrive at a contradiction.

Example

Given: $x \in \left[0, \frac{\pi}{2}\right]$
Claim: $\sin x + \cos x \geq 1$.
Proof: Assume that $\forall x \in \left[0, \frac{\pi}{2}\right] : \sin x + \cos x < 1$.
For $x \in \left[0, \frac{\pi}{2}\right]$, we have $\sin x \geq 0$ and $\cos x \geq 0$.

$$\Rightarrow 0 \leq \sin x + \cos x < 1 \Rightarrow 0^2 \leq (\sin x + \cos x)^2 < 1^2$$

$$\Rightarrow 0^2 \leq \sin^2 x + \cos^2 x + 2 \sin x \cos x = 1 + 2 \sin x \cos x < 1 \Rightarrow \sin x \cos x < 0$$

which contradicts the fact that both terms are non-negative since the product must be non-negative.

Exercise

1. Prove that $\sqrt{6}$ is irrational.
2. If A and B are sets, then $A \cap (B - A) = \emptyset$.

Outline

1. Mathematical Logic

2. Proof Techniques

2. 5 Proof by exhaustion

Proof by exhaustion

In proof by exhaustion, the conclusion is established by dividing it into a finite number of cases and proving each one separately. The number of cases sometimes can become very large.

Example

Given: $n \in \mathbb{Z}$.
Claim: n is even if and only if n^2 is even.
Proof: We consider two cases:
<u>Case 1: n is even</u>
If n is even, then $n = 2k$ for some integer k. Thus, $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$ Since $2k^2$ is an integer, n^2 is even.
<u>Case 2: n is odd</u>
If n is odd, then $n = 2k + 1$ for some integer k. Thus, $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ Since $2k^2 + 2k$ is an integer, n^2 is odd.
Therefore, an integer n is even if and only if n^2 is even.

Outline

1. Mathematical Logic

2. Proof Techniques

2. 6 Proofs involving sets

2. 6. 1 How to prove $a \in A$?

2. 6. 2 How to prove $A \subseteq B$?

2. 6. 3 How to prove $A = B$?

Proofs involving sets

How to prove $a \in A$?

Rewrite the set A in set-builder notation with P being the property that is required for *membership* of set A:

1. $A = \{x : P(x)\}$
   Show that $P(a)$ is true.
2. $A = \{x \in S : P(x)\}$
   First, verify that $a \in S$, then show that $P(a)$ is true.

How to prove $A \subseteq B$?

If A and B are sets, then $A \subseteq B$ means that every element of A is also an element of B. The proof reduces to showing that: If $a \in A$, then $a \in B$.

Example: Given that 2^k is the set of all subsets of k. If A and B are sets, then $2^A \cup 2^B \subseteq 2^{A \cup B}$.

Exercise: Suppose A and B are sets. If $2^A \subseteq 2^B$, then $A \subseteq B$.

How to prove $A = B$?

If A and B are sets, then $A = B$ means that every element of A is also an element of B and every element of B is also an element of A. The proof reduces to showing that: $A \subseteq B$ and $B \subseteq A$.

Example: If A, B, C are sets, then $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Exercise: Suppose A, B, C are sets. Prove that if $A \subseteq B$ then $A - C \subseteq B - C$.